Notes: Mehra & Prescott (JME, 1985)

The Equity Premium: A Puzzle

Contents

1 Big picture

- **Question.** Can a frictionless Arrow–Debreu/Lucas-style representative-agent economy explain both the high average U.S. equity return and the low average risk-free return?
- **Empirical target.** In 1889–1978 U.S. data, the average real annual return on the S&P 500 is about 6.98%, the average real return on short-term riskless debt is about 0.80%, and the average equity premium is about 6.18%.
- **Model class.** The paper studies a pure exchange, representative-agent economy with CRRA utility and a stationary Markov process for consumption growth.
- **Calibration discipline.** The consumption-growth process is chosen to match the U.S. mean, variance, and first-order serial correlation of real per capita consumption growth.
- **Main finding.** For plausible values of risk aversion and time discounting, the model cannot generate anything close to the observed equity premium while also keeping the risk-free rate low.
- **Key numerical result.** In the model's admissible region with average real risk-free rates between 0 and 4%, the maximum average equity premium is only about 0.35–0.40%, far below the observed 6.18%.
- **Economic intuition.** High risk aversion can raise the equity premium, but in a growing economy high curvature also makes the representative agent want to shift consumption toward the present, pushing the risk-free rate far above the data.
- **Robustness.** The result is robust to measurement errors, tax considerations, aggregation, alternative consumption-growth processes, firm leverage, and adding production in the same Arrow–Debreu spirit.
- **Conclusion.** The puzzle is not only why equity returns are high. It is also why safe real rates are so low. The authors argue that resolving both facts likely requires a model outside the frictionless Arrow–Debreu framework.

2 Data and measurement

2.1 Sample period and basic series

- The sample is annual U.S. data from 1889 to 1978.
- The paper uses five basic series: real stock prices, real dividends, real per capita consumption, the consumption deflator, and nominal yields on short-term relatively riskless securities.
- The stock return is based on the Standard & Poor's composite index plus dividends.
- The consumption series is real per capita consumption on nondurables and services.
- The risk-free return series uses Treasury bills after 1931, Treasury certificates for 1920–1930, and 60-to-90-day prime commercial paper before 1920.

Interpretation: The data are designed to compare a broad equity claim with a short-term default-free or nearly default-free safe asset, while disciplining the model with the observed consumption process.

2.2 Equity return

Let P_t denote the real annual average S&P composite stock price and D_t real dividends. The real return on equity in year t is

$$R_t^e = \frac{P_{t+1} + D_t - P_t}{P_t}. \tag{1}$$

Interpretation: This is the real total return to holding the market equity claim for one year.

2.3 Real risk-free return

Let RF_t be the nominal yield on the short-term riskless security and let PC_t be the consumption deflator. The real risk-free return is calculated as

$$R_t^f = RF_t - \frac{PC_{t+1} - PC_t}{PC_t}. \tag{2}$$

Interpretation: The paper treats this as the realized real return on a short-term nominally safe asset. Measurement error in inflation affects both safe and equity real returns, so it is not enough to close the premium gap.

2.4 Equity premium

The annual risk premium is

$$RP_t = R_t^e - R_t^f. \tag{3}$$

The sample mean premium is

$$\overline{RP} = \overline{R^e} - \overline{R^f} = 6.98\% - 0.80\% = 6.18\%. \tag{4}$$

Interpretation: The empirical target is large not only in mean but also relative to what consumption risk in a representative-agent model can support.

2.5 Consumption-growth targets

The model's consumption-growth process is chosen to match:

- mean growth of real per capita consumption: about 1.8% per year;
- standard deviation of consumption growth: about 3.6% per year;
- first-order serial correlation: about -0.14.

Interpretation: Consumption growth is much smoother than equity returns. This smoothness is what makes the representative-agent explanation of the equity premium difficult.

3 Models and empirical specifications

3.1 Representative-agent preferences

The representative household has expected utility

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t U(c_t) \right], \qquad 0 < \beta < 1. \tag{5}$$

Utility is constant relative risk aversion:

$$U(c, \alpha) = \frac{c^{1-\alpha} - 1}{1 - \alpha}, \qquad 0 < \alpha < \infty, \tag{6}$$

with log utility as the limit when $\alpha = 1$.

Interpretation: The parameter α is the curvature of utility and is inversely related to intertemporal substitution. Larger α makes consumption risk more costly, but also raises the risk-free rate in a growing economy.

3.2 Endowment growth

The economy has one productive unit. Its output equals the dividend and consumption endowment. The growth rate follows a Markov chain:

$$y_{t+1} = x_{t+1} y_t, \tag{7}$$

where

$$x_{t+1} \in \{\lambda_1, \dots, \lambda_n\}, \qquad \Pr(x_{t+1} = \lambda_j \mid x_t = \lambda_i) = \phi_{ij}. \tag{8}$$

Interpretation: The paper makes consumption growth stationary rather than the level of consumption. This accommodates long-run growth while keeping returns stationary.

3.3 General asset-pricing formula

For any security with dividend process $\{d_s\}$, the ex-dividend price at date t is

$$P_t = E_t \left[\sum_{s=t+1}^{\infty} \beta^{s-t} \frac{U'(y_s)}{U'(y_t)} d_s \right]. \tag{9}$$

Interpretation: Asset prices are discounted expected payoffs using the representative agent's intertemporal marginal rate of substitution.

3.4 Equity price

For the equity claim, dividends equal aggregate output. If current state is (c, i), the equity price satisfies

$$p^e(c, i) = \beta \sum_{j=1}^{n} \phi_{ij} (\lambda_j c)^{-\alpha} \left[p^e(\lambda_j c, j) + \lambda_j c \right] c^{\alpha}. \tag{10}$$

Because the equity price is homogeneous of degree one in current output,

$$p^e(c, i) = w_i c. \tag{11}$$

Substitution gives the linear system

$$w_i = \beta \sum_{j=1}^{n} \phi_{ij} \lambda_j^{1-\alpha} (w_j + 1), \qquad i = 1, \dots, n. \tag{12}$$

Interpretation: The constants w_i are price-dividend ratios by state. They summarize how expected growth and risk premia vary with the current consumption-growth state.

3.5 Equity return

If current state is i and next state is j, the equity return is

$$r_{ij}^e = \frac{p^e(\lambda_j c, j) + \lambda_j c - p^e(c, i)}{p^e(c, i)} = \frac{\lambda_j (w_j + 1)}{w_i} - 1. \tag{13}$$

The expected equity return conditional on current state i is

$$R_i^e = \sum_{j=1}^{n} \phi_{ij} r_{ij}^e. \tag{14}$$

Interpretation: Equity is risky because its payoff covaries with consumption growth. But observed consumption fluctuations are too small to generate the historical premium under plausible curvature.

3.6 Risk-free bill price and return

A one-period real bill pays one unit of consumption next period. Its price is

$$p_i^f = \beta \sum_{j=1}^{n} \phi_{ij} \lambda_j^{-\alpha}. \tag{15}$$

The certain one-period return is

$$R_i^f = \frac{1}{p_i^f} - 1. \tag{16}$$

Interpretation: The risk-free return is pinned down by expected marginal utility growth. With positive consumption growth, high α lowers expected future marginal utility and raises the safe real rate.

3.7 Stationary average returns

Let π_i be the stationary distribution of the Markov chain. Then

$$R^e = \sum_{i=1}^{n} \pi_i R_i^e, \qquad R^f = \sum_{i=1}^{n} \pi_i R_i^f. \tag{17}$$

The model-implied equity premium is

$$R^e - R^f. \tag{18}$$

Interpretation: These are the theoretical counterparts of the sample average equity return, risk-free return, and equity premium.

3.8 Two-state parameterization

The paper uses two states:

$$\lambda_1 = 1 + \mu + \delta, \qquad \lambda_2 = 1 + \mu - \delta, \tag{19}$$

with transition probabilities

$$\phi_{11} = \phi_{22} = \phi, \qquad \phi_{12} = \phi_{21} = 1 - \phi. \tag{20}$$

The values are chosen to match consumption-growth moments:

$$\mu = 0.018, \qquad \delta = 0.036, \qquad \phi = 0.43. \tag{21}$$

Interpretation: This parameterization independently controls the mean, volatility, and serial correlation of consumption growth.

3.9 Admissible preferences

The paper searches over α and β subject to plausible restrictions:

- α is restricted to a range consistent with micro, macro, and international evidence;
- $0 < \beta < 1$;
- the implied average risk-free return is constrained to a plausible range, especially 0–4%.

Interpretation: High α can mechanically raise the equity premium, but the same high α usually produces an implausibly high safe real rate. The puzzle is the joint failure.

3.10 Firm leverage extension

The model equity claim initially corresponds to all output. To make equity more like corporate stock, the paper considers a claim whose dividend is output net of a fixed commitment. Let θ be the fraction of expected next-period output committed at date t. The equity-pricing equation becomes

$$p^e(c,i) = \beta \sum_{j=1}^{n} \phi_{ij} (\lambda_j c)^{-\alpha} \left[p^e(\lambda_j c, j) + c\lambda_j - \theta \sum_{k=1}^{n} \phi_{ik} c \lambda_k \right] c^{\alpha}. \tag{22}$$

Interpretation: Setting $\theta = 0.9$ makes equity receive roughly 10% of output, resembling the corporate profit share. This barely changes the result: it raises the equity premium by less than one-tenth of one percent.

4 Identification and empirical design

4.1 What identifies the test

- This is not a regression or parameter-estimation paper. It is a quantitative theoretical test.
- The model is disciplined by the historical consumption-growth process and by plausible preference bounds.
- The question is whether any economy in this model class can jointly match the observed safe rate and equity premium.

Interpretation: The test is powerful because it gives the model flexibility over preferences while holding it to the observed consumption process.

4.2 Why the consumption process is central

- In the model, aggregate consumption is the sole source of systematic risk.
- Equity is risky because it pays off in states tied to aggregate consumption.
- If consumption growth is smooth, the representative agent does not require a huge premium to bear equity risk unless risk aversion is very high.

Interpretation: The empirical smoothness of consumption is the core reason that the premium is hard to explain.

4.3 Why the risk-free rate restriction matters

- Raising risk aversion increases the equity premium.
- But in a growing economy, high risk aversion also implies low marginal utility in the future and therefore a high desire to borrow against future consumption.
- Equilibrium clears this desire through a high real risk-free rate.

Interpretation: The model can generate a high premium or a low safe rate, but not both at the same time.

4.4 Robustness arguments

The authors consider several possible objections:

- inflation measurement errors;
- taxes and after-tax returns;

- aggregation frequency;
- alternative distributions of consumption growth;
- firm leverage and fixed payments to non-equity claimants;
- production or capital accumulation variants.

Interpretation: These variations do not overturn the conclusion. The gap is too large to be closed by small specification changes.

4.5 What the paper cannot fully prove

- It does not identify the unique missing friction.
- It does not claim no Arrow–Debreu model can ever match every feature under all possible preference assumptions.
- It does not solve price volatility or predictability puzzles.
- It abstracts from heterogeneity, incomplete markets, liquidity constraints, transaction costs, borrowing constraints, rare disasters, and money-like convenience yields.

Interpretation: The paper's contribution is to isolate a precise failure of the canonical frictionless model, not to provide the final alternative model.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format, while grouping adjacent original visuals to save space. The table and figures are directly cropped from the original paper.

5.1 Table 1 + Figure 4: the observed puzzle versus the model's admissible region

Read:

- Table 1 reports the core historical facts. Over 1889–1978, real per capita consumption growth averages 1.83% with a standard deviation of 3.57%.
- The average real return on relatively riskless securities is only 0.80%, while the S&P 500 real return is 6.98%.
- The average risk premium is therefore 6.18% with a standard error of 1.76%.
- Figure 4 plots the model's admissible combinations of average risk-free rates and average equity premia.
- The admissible region sits around risk-free rates of roughly 2–4% and equity premia below about 0.4%.
- The observed combination, risk-free rate near 0.8% and premium near 6%, lies far outside the model's region.

Economic message: The model does not miss by a little. It misses by an order of magnitude on the equity premium, while also struggling with the low risk-free rate.

Table 1

Time periods	% growth rate of per capita real consumption: Mean	Standard deviation	% real return on a relatively riskless security: Mean	Standard deviation	% risk premium: Mean	Standard deviation	% real return on S&P 500: Mean	Standard deviation
1889–1978	1.83 (Std error = 0.38)	3.57	0.80 (Std error = 0.60)	5.67	6.18 (Std error = 1.76)	16.67	6.98 (Std error = 1.74)	16.54
1889–1898	2.30	4.90	5.80	3.23	1.78	11.57	7.58	10.02
1899–1908	2.55	5.31	2.62	2.59	5.08	16.86	7.71	17.21
1909–1918	0.44	3.07	−1.63	9.02	1.49	9.18	−0.14	12.81
1919–1928	3.00	3.97	4.30	6.61	14.64	15.94	18.94	16.18
1929–1938	−0.25	5.28	2.39	6.50	0.18	31.63	2.56	27.90
1939–1948	2.19	2.52	−5.82	4.05	8.89	14.23	3.07	14.67
1949–1958	1.48	1.00	−0.81	1.89	18.30	13.20	17.49	13.08
1959–1968	2.37	1.00	1.07	0.64	4.50	10.17	5.58	10.59
1969–1978	2.41	1.40	−0.72	2.06	0.75	11.64	0.03	13.11

(a) Table 1: historical summary statistics.

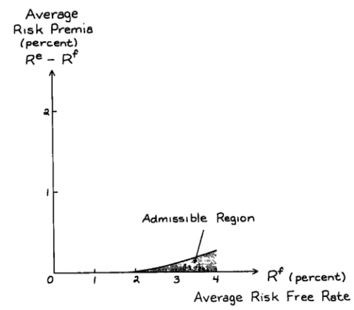

Fig. 4. Set of admissible average equity risk premia and real returns.

(b) Figure 4: admissible risk-free rates and premia.

5.2 Figures 1–3: the raw time series behind the puzzle

Read:

- Figure 1 shows that real equity returns are highly volatile, with large positive and negative episodes.
- Figure 2 shows that consumption growth is much smoother than equity returns.
- Figure 3 shows that the real return on the short-term riskless security is low on average, with large disruptions mostly around inflationary or deflationary episodes.
- The mismatch between Figure 1 and Figure 2 is central: equity returns are volatile, but consumption risk is not large enough to justify the observed premium in the model.

Economic message: The puzzle is visible before any model is written down: the representative-agent model prices risk using consumption, but consumption growth is too smooth to explain the equity-return spread.

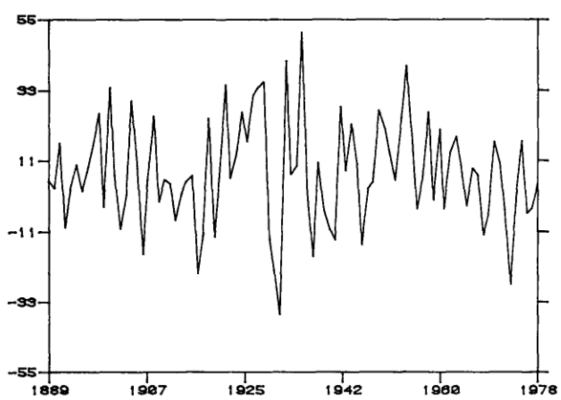

Fig. 1. Real annual return on S&P 500, 1889–1978 (percent).

(a) Figure 1: real annual S&P 500 returns.

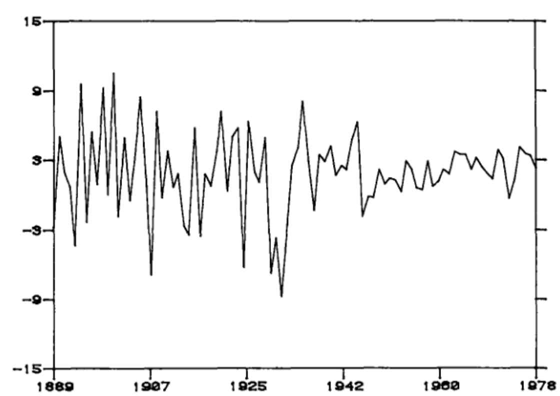

Fig. 2. Growth rate of real per capita consumption, 1889–1978 (percent).

(b) Figure 2: real per capita consumption growth.

Figure 3: Real return on relatively riskless securities.

6 Short summary

- The paper documents the classic U.S. equity premium puzzle for 1889–1978.
- The historical equity premium is about 6.18% per year.
- The average risk-free real rate is only about 0.80% per year.
- Real per capita consumption growth averages about 1.83% with a standard deviation of 3.57%.
- The model is a Lucas-style pure exchange economy with a representative agent, CRRA utility, and Markov consumption growth.
- Equity is a claim to aggregate output, and a one-period real bill is the safe asset.
- Asset prices are determined by expected marginal utility growth.
- The consumption-growth process is matched to the U.S. mean, variance, and serial correlation.
- The authors search over plausible risk aversion and discount factors.
- The model's maximum equity premium, while keeping the risk-free rate plausible, is only about 0.35–0.40%.
- High risk aversion can raise the premium, but it also makes the risk-free rate too high.
- Leverage-like modifications to the equity claim do not solve the puzzle.
- Measurement, tax, and process-misspecification concerns also do not solve it.
- The paper concludes that a successful explanation likely requires frictions or departures from the Arrow–Debreu representative-agent framework.

7 Conclusion

7.1 Core conclusion

Mehra and Prescott show that a standard frictionless representative-agent model cannot jointly explain the high average return on equity and the low average return on short-term safe debt. Matching consumption growth and allowing plausible preference parameters leaves the model far from the observed equity premium.

7.2 Economic intuition

The representative agent prices equity risk through aggregate consumption risk. But aggregate consumption growth is not very volatile. To make the agent demand a 6% equity premium, one must make marginal utility extremely sensitive to consumption. In a growing economy, that same sensitivity makes future marginal utility low relative to current marginal utility, which pushes the equilibrium risk-free rate too high.

7.3 Takeaway

The paper's enduring lesson is that asset pricing is constrained jointly by risky returns and safe returns. Explaining the equity premium alone is not enough; a successful theory must also explain why the safe real rate is so low.